

Month 1 - Foundations: Python & Math for ML

Topics:

- Python Fundamentals (Data Types, Loops, Functions, OOP basics)
- NumPy for Numerical Operations
- Pandas for Data Manipulation
- Basic Probability & Statistics (Mean, Median, Variance, Distributions)
- Linear Algebra Basics (Vectors, Matrices, Operations)

Project:

- **Data Exploration and Cleaning Tool**: Use Pandas to load a CSV dataset, handle missing values, and perform basic descriptive statistics.

Resources:

- [Python for Everybody (Free Course)](<https://www.py4e.com/>)
- [NumPy Official Documentation](<https://numpy.org/doc/stable/>)
- [Pandas Official Documentation](<https://pandas.pydata.org/docs/>)
- [Khan Academy Statistics and Probability](<https://www.khanacademy.org/math/statistics-probability>)
- [Khan Academy Linear Algebra](<https://www.khanacademy.org/math/linear-algebra>)

Practice:

- [HackerRank (Python)](<https://www.hackerrank.com/domains/python>)
- [LeetCode (Easy problems focusing on data structures/algorithms)](<https://leetcode.com/problemset/all/>)

Month 2 - Core Machine Learning Concepts

Topics:

- Introduction to Machine Learning (Types of ML: Supervised, Unsupervised, Reinforcement)
- Supervised Learning:
 - Linear Regression
 - Logistic Regression
 - Decision Trees
 - Introduction to Model Evaluation (Accuracy, Precision, Recall, F1-Score)
- Unsupervised Learning:
 - K-Means Clustering
- Scikit-learn Library Introduction

Project:

- **Predicting House Prices**: Build a Linear Regression model using Scikit-learn to predict house prices based on features from a dataset (e.g., Boston Housing Dataset).

Resources:

- [Scikit-learn User Guide](https://scikit-learn.org/stable/user_guide.html)
- [Machine Learning Crash Course with TensorFlow APIs (Google)](<https://developers.google.com/machine-learning/crash-course>)
- [Introduction to Machine Learning with Python (Book - freely available online)](<https://www.IntroToMLWithPython.com/>)

Practice:

- [Kaggle (Beginner Competitions & Datasets)](<https://www.kaggle.com/>)
- [HackerRank (SQL - useful for data handling)](<https://www.hackerrank.com/domains/sql>)

Month 3 - Deep Dive & Interview Prep

Topics:

- Advanced Supervised Learning: Support Vector Machines (SVMs), Random Forests
- Ensemble Methods: Bagging and Boosting
- Neural Networks Basics (Perceptrons, Activation Functions, Backpropagation)
- Introduction to Deep Learning Frameworks (TensorFlow/PyTorch - basic usage)
- Overfitting and Regularization Techniques
- Feature Engineering and Selection
- Interview Preparation (Common ML interview questions, problem-solving approach)

Project:

- **Image Classification**: Train a simple Neural Network or a Random Forest model to classify images from a dataset like MNIST. Focus on the entire pipeline from data loading to model evaluation.

Resources:

- [Deep Learning Specialization by Andrew Ng (Coursera - audit for free)](<https://www.coursera.org/specializations/deep-learning>)
- [TensorFlow Tutorials](<https://www.tensorflow.org/tutorials>)
- [PyTorch Tutorials](<https://pytorch.org/tutorials/>)
- [Elements of AI (Free Course)](<https://www.elementsofai.com/>)

Practice:

- [LeetCode (Medium problems & ML-specific questions if available)](<https://leetcode.com/problemset/all/>)
- [InterviewBit (ML Interview

Questions)](<https://www.interviewbit.com/machine-learning-interview-questions/>)